

Salomon Kabongo, PhD

847-912-8096 | kabongosalomon@gmail.com | linkedin.com/in/salomon-kabongo | github.com/Kabongosalomon

Normal, IL | US Permanent Resident (Green Card Holder) – No Sponsorship Required

EDUCATION

Leibniz Universität Hannover

PhD in Computer Science, AI - Natural Language Processing (LLMs)

Hannover, Germany

Nov. 2020 – Nov. 2025

African Master's in Machine Intelligence (AMMI)

Master's in Machine Intelligence

Accra, Ghana

Oct. 2019 – Nov. 2020

University of the Western Cape

Master's in Mathematical Sciences

Cape Town, South Africa

Aug. 2018 – Jun. 2019

Université de Lubumbashi

Bachelor of Science (Honours) in Mathematics and Computer Science

Lubumbashi, DRC

Oct. 2014 – Jul. 2017

EXPERIENCE

Lead Software Engineer

State Farm - Innovation Group

Feb. 2022 – Present

Bloomington-Normal, IL

- Designed automated pre-labeling pipelines using embedding-based retrieval to accelerate data annotation.
- Architected a proprietary document deduplication system utilizing visual similarity and hashing algorithms to identify near-duplicates, significantly streamlining business workflows.
- Led R&D initiatives on Synthetic Media (Deepfake) detection and Video Understanding; benchmarked Visual Language Models (VLMs) against vendor solutions to inform strategic technology procurement.
- Invented novel computer vision applications for the insurance domain, resulting in **4 patent filings (pending approval)**.

Board Member

Masakhane Research Foundation

2021 – Present

Global

- Spearheaded the strategic formation of the **Masakhane AI Hub**, defining the 2025–2029 roadmap to build digital public infrastructure for 1 billion+ African language speakers.
- Secured and currently overseeing the execution of **~\$9M USD in research funding** (including \$5M from the Bill & Melinda Gates Foundation and \$4M from IDRC) to democratize AI access.
- Direct high-level collaborations with strategic partners including **Google.org**, **Lacuna Fund**, and **UNESCO**, scaling the community's impact across 50+ African languages.

Research Assistant

L3S/Leibniz Information Center for Science & Technology (TIB)

Nov. 2020 – Nov. 2022

Hannover, Germany

- Engineered the core "Leaderboards" feature for the Open Research Knowledge Graph (ORKG), utilizing Knowledge Graphs to automatically track and visualize state-of-the-art (SOTA) progress across scientific publications.
- Collaborated with Hannover Medical School (MHH) on personalized medicine research, applying machine learning techniques to analyze large-scale genetic datasets for predictive healthcare outcomes.
- Conducted research on Scholarly Information Extraction, developing novel NLP pipelines to extract metric data from unstructured text for knowledge graph construction.

SELECTED PUBLICATIONS & PATENTS

Systems and Methods for Advanced Duplicate Image Search and Analysis <i>US Patent</i>	2024
• US Patent App. 18/652,500, Publication No. US20240411724A1: Invented a system for identifying duplicate documents using vector embeddings and similarity hashing (Assignee: State Farm).	
Bibletts & LiSTra: African Speech Corpora <i>Interspeech, NeurIPS Workshops</i>	2022
• Co-authored "Bibletts", a high-fidelity multilingual speech corpus. • Developed LiSTra, the first English-to-Lingala speech translation baseline.	
Automated Mining of Leaderboards for Empirical AI Research <i>ICADL Best Paper Award</i>	2021
• Proposed a systematic workflow to extract SOTA performance metrics from unstructured scientific text. • Published in ICADL and International Journal on Digital Libraries (30+ citations).	
Participatory Research for Low-Resourced Machine Translation <i>EMNLP Findings</i>	2020
• Contributor to the Masakhane NLP initiative; paper has over 280+ citations and set the standard for African Language NLP. • Developed benchmarks and datasets that are now standard in the field.	

TECHNICAL SKILLS

Languages: Python, C/C++, SQL, Bash
Deep Learning & AI: PyTorch, TensorFlow, Hugging Face Transformers, LLMs, RAG, LangChain, OpenCV
Cloud & MLOps: AWS (SageMaker, Lambda), Docker, Kubernetes, Google Cloud Vertex AI, Linux, Git
Research Areas: NLP, Computer Vision, Knowledge Graphs, Speech Translation, Generative AI